

Predicting Falcon 9 First-Stage Landing Success

Applied Data Science Capstone Project

Executive Summary

Objective

Predict whether the Falcon 9 first stage will successfully land after launch.

Data & Methods

- Collected launch data using the SpaceX API and web scraping
- Cleaned and transformed the data for analysis
- Explored launch outcomes using Python visualizations and SQL
- Analyzed launch sites and proximity using Folium
- Developed an interactive dashboard using Plotly Dash
- Trained and evaluated Logistic Regression, SVM, Decision Tree, and KNN models

Key Findings

- The analysis dataset contains 90 Falcon 9 launches
- Overall landing success rate: 66.67%
- CCAFS SLC-40 recorded the largest number of launches
- Landing outcomes varied across launch sites, payloads, and orbits
- Logistic Regression, SVM, and KNN each achieved 83.33% test accuracy
- Decision Tree achieved 66.67% test accuracy

Introduction

Background

SpaceX has significantly reduced the cost of space launches through the development and reuse of rocket hardware.

The Falcon 9 first-stage booster is designed to return to Earth after launch and attempt a controlled landing for future reuse.

Problem

A successful landing depends on multiple factors, including:

- Launch site
- Payload mass
- Orbit
- Flight history
- Booster and landing characteristics

Project Objective

Use historical Falcon 9 launch data to:

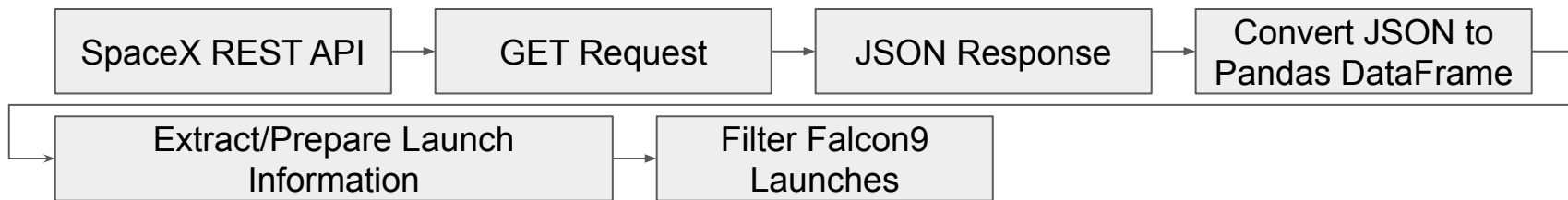
1. Explore patterns associated with first-stage landing success.
2. Visualize launch and landing outcomes.
3. Analyze the data using SQL and interactive visualization.
4. Build machine-learning classifiers to predict landing success.

Data Collection — SpaceX API

Objective

Collect historical Falcon 9 launch data from the SpaceX API for analysis and prediction.

API Data Collection Process



Key Data

- Launch date and time
- Launch site
- Payload information
- Orbit
- Booster version
- Landing outcome
- Landing pad

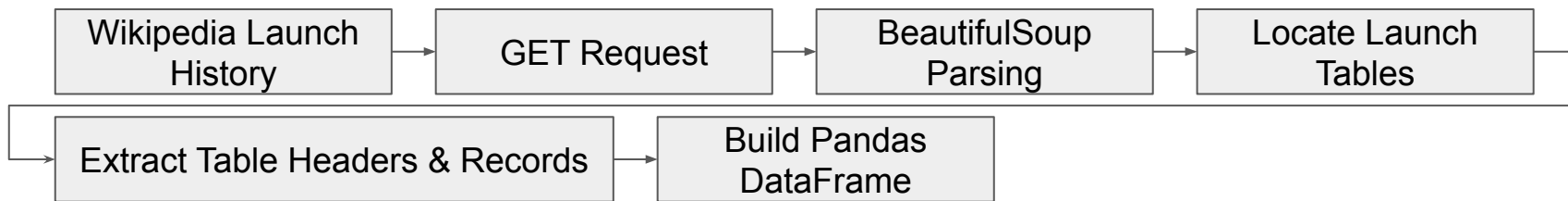
Source: SpaceX API / IBM Skills Network dataset

Data Collection — Web Scrapping

Objective

Collect additional Falcon 9 launch records from the Wikipedia launch history.

Web Scrapping Process



Extracted information

- Launch date and time
- Launch site
- Payload information
- Orbit
- Booster version
- Landing outcome
- Landing pad

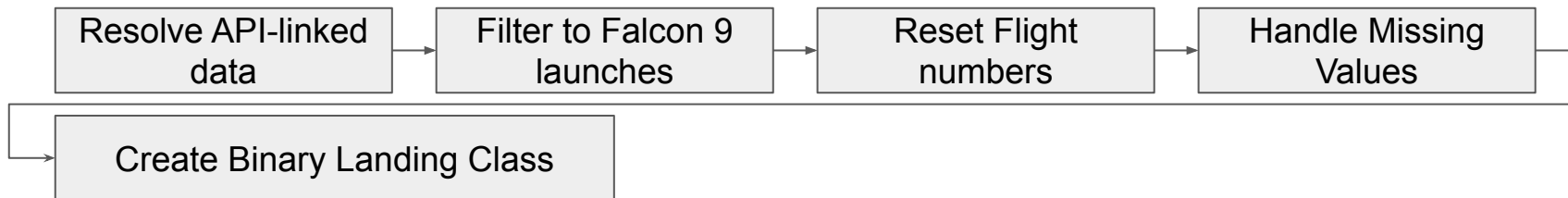
Source: Wikipedia — List of Falcon 9 and Falcon Heavy launches

Data Wrangling

Objective

Prepare the Falcon 9 dataset for exploratory analysis and machine learning.

Data Processing



Key Transformations

- Extracted booster, payload, orbit, launch-site and core information
- Converted launch dates to date format
- Retained None in LandingPad where no landing pad was used
- Reset Flight numbers
- Created Class: 1 = Successful landing, 0 = Unsuccessful landing
- Exported the cleaned dataset for subsequent analysis

Result

90 Falcon 9 launches

66.67% overall landing success rate

EDA - Data Visualization

Objective

Explore relationships between launch characteristics and first-stage landing success.

Visualizations Used

Launch Site

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site

Orbit

- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit

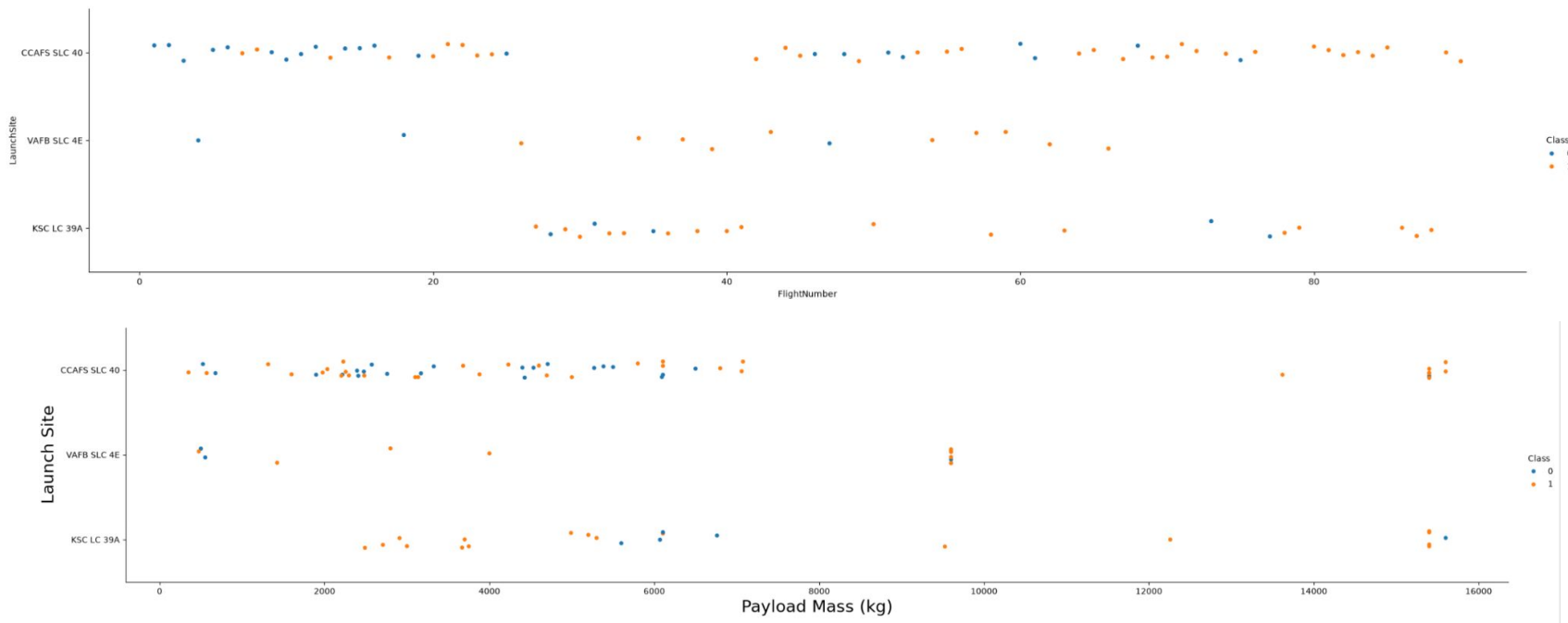
Time

- Yearly Landing Success Trend

Analysis Focus

- Compare successful and unsuccessful landing outcomes
- Examine how payload mass varies across launch sites and orbits
- Identify changes in landing success over time
- Explore patterns that can inform predictive modeling

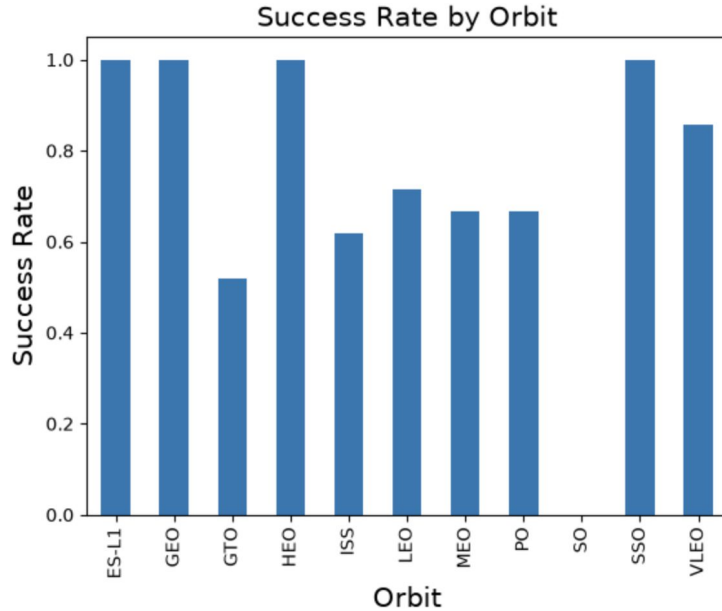
EDA Results: Launch Site Analysis



Key Observations

- CCAFS SLC 40 accounts for the largest number of launches in the dataset
- The dataset contains launches from three launch sites
- Payload masses vary across launch sites, with substantial overlap between sites

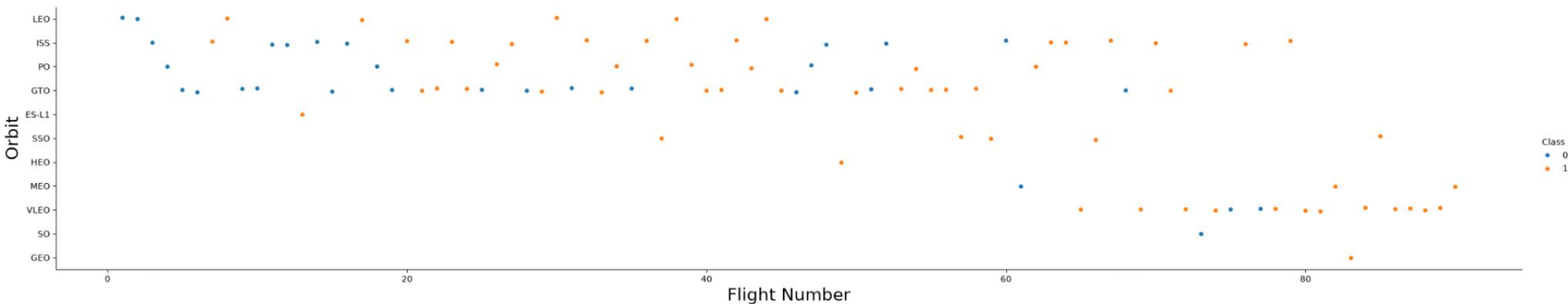
EDA Results: Success Rate by Orbit



Key Observations

- Success rates vary substantially across orbit types.
- ES-L1, GEO, HEO, and SSO show 100% landing success in the dataset.
- GTO has the lowest non-zero success rate at approximately 52%.
- VLEO shows a relatively high success rate of approximately 86%.

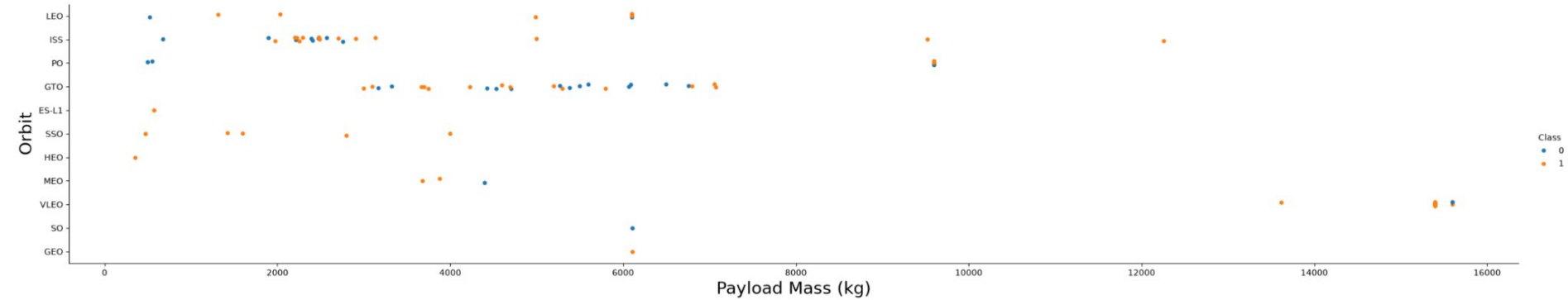
EDA Results: Flight Number vs Orbit



Key Observations

- In LEO, successful landings appear more frequently at higher flight numbers.
- In GTO, successful and unsuccessful outcomes are mixed, with no clear relationship between flight number and landing success.
- This suggests that landing outcomes vary by orbit and flight history in the analysed dataset.

EDA Results: Payload Mass vs Orbit



Key Observations

- For Polar, LEO, and ISS, successful landings are observed across heavier payload ranges.
- In GTO, successful and unsuccessful outcomes occur across the payload range, making the relationship less distinct.
- Landing success therefore appears to vary depending on both payload mass and orbit.

EDA Results: Yearly Landing Success Trend



Key Observations

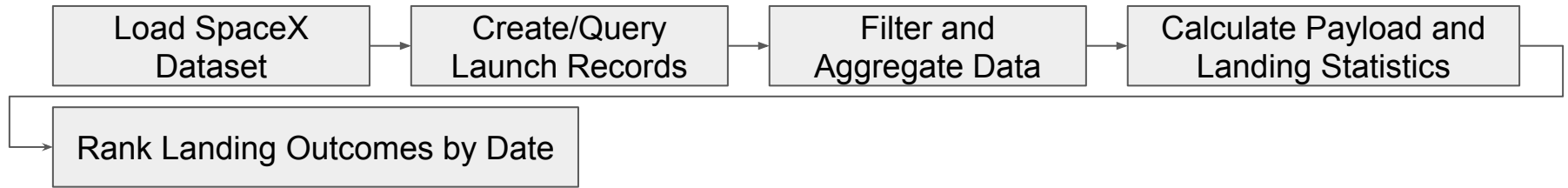
- Landing success generally increases across the later years of the dataset.
- The trend shows noticeable year-to-year variation rather than a constant success rate.
- Landing success reaches high levels in the later years, particularly around 2019–2020.

SQL Analysis Methodology

Objective

Use SQL queries to examine launch sites, payloads, landing outcomes, and historical landing performance

Process



SQL Analysis Areas

- Launch site identification and filtering
- Payload mass calculations
- Landing outcome counts
- Booster versions and payload capacity
- Historical landing outcomes and date-based analysis

SQL Results: Launch Sites & Payload Analysis

Key Observations

- First successful ground-pad landing: 22 December 2015
- This marks the earliest recorded successful ground-pad landing in the SQL analysis dataset

Launch-site analysis

- Unique launch sites
- Launch records beginning with CCA

Payload analysis

- Total payload mass for NASA CRS missions: 48,213 kg
- Average payload mass for Falcon 9 v1.1: 2,534.67 kg

SQL Results: Landing Outcomes & Historical Analysis

Landing outcome analysis

- Mission outcome counts: 98 successful missions, 1 in-flight failure, plus 2 differently formatted success categories in the SQL dataset
- Successful drone-ship landings with 4,000–6,000 kg payload: 4 records
- 2015 drone-ship landing failures: 2 records - January (F9 v1.1 B1012, CCAFS LC-40) and April (F9 v1.1 B1015, CCAFS LC-40)

Booster & historical analysis

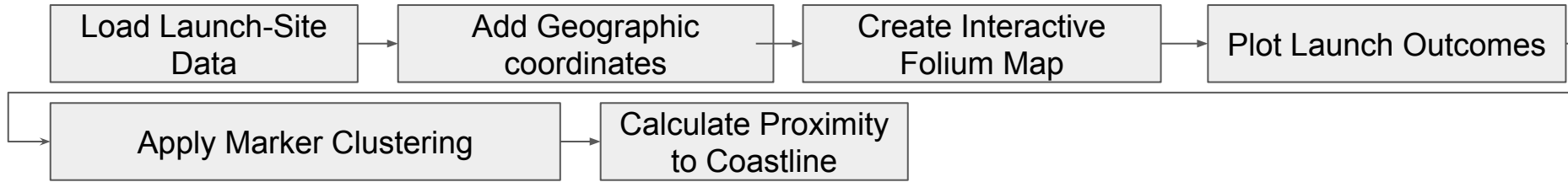
- 12 booster records are tied for the maximum payload carried.
- Ranked landing outcomes (2010–2017) show multiple outcome categories, including successful and failed drone-ship landings, ground-pad successes, and ocean outcomes.

Interactive Visual Analytics: Folium

Objective

Visualize Falcon 9 launch sites and landing outcomes geographically

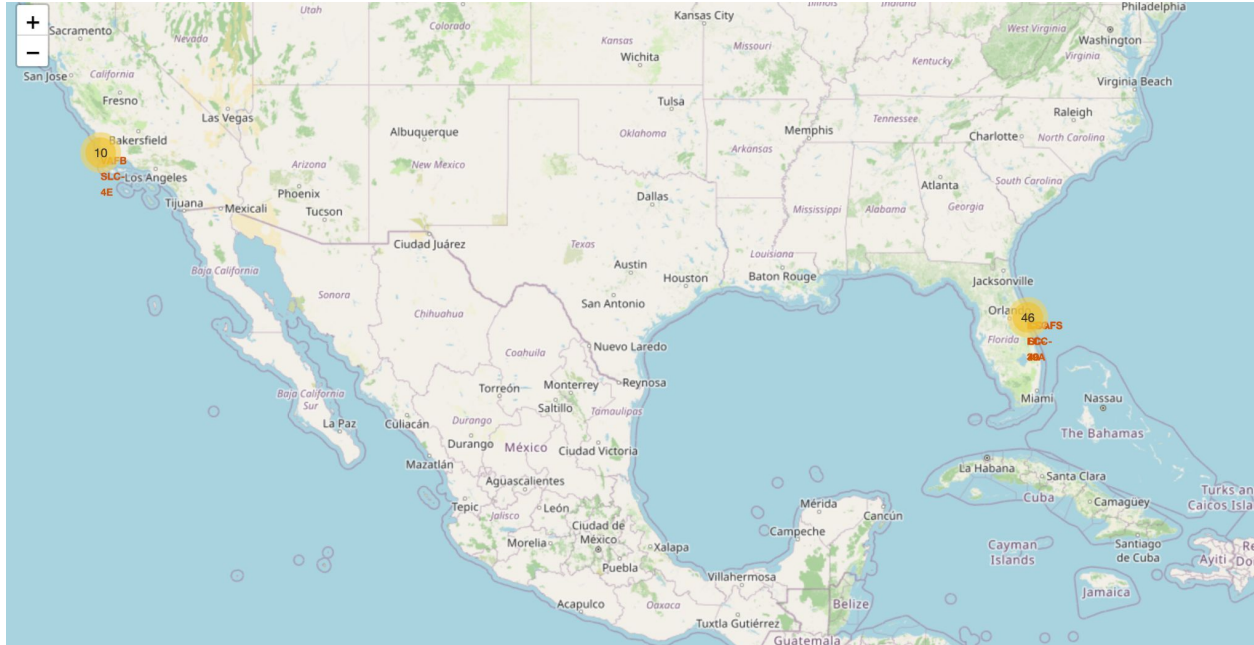
Process



Analysis

- Mapped Falcon 9 launch sites using geographic coordinates
- Displayed successful and unsuccessful launches
- Grouped launch records using marker clustering
- Analyzed launch-site proximity to the coastline

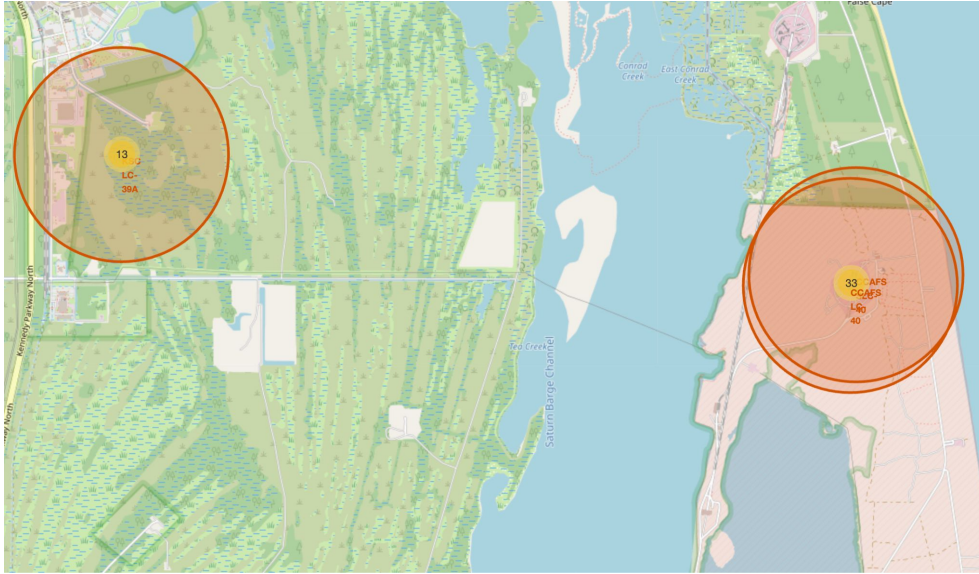
Interactive Visual Analytics: Folium - Launch Sites



Geographic Visualization

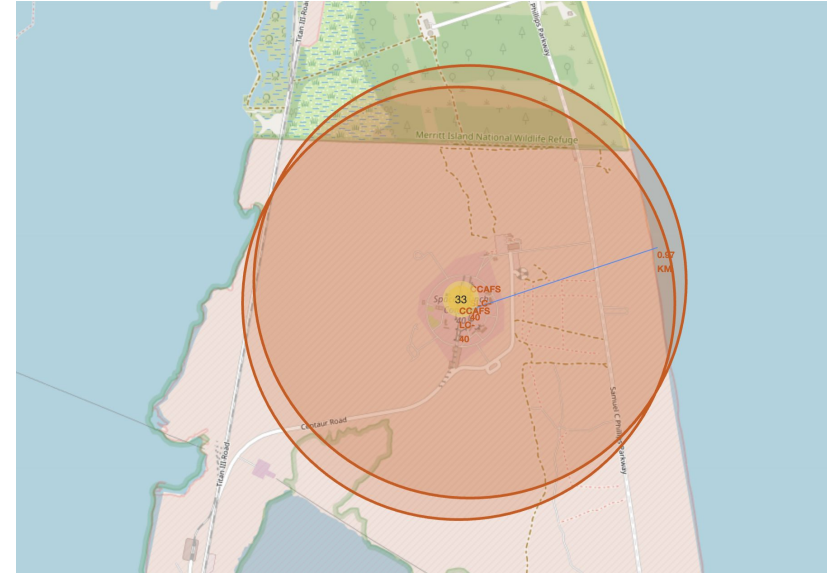
- 4 launch-site locations were represented using geographic coordinates
- Launch sites were displayed on an interactive map using Folium markers and circles
- The map provides a geographic view of the Falcon 9 launch locations

Interactive Visual Analytics: Folium - Landing Outcomes



Landing Outcomes

- Successful and unsuccessful launches displayed geographically
- Marker clustering used to group launch records



Coastline Proximity

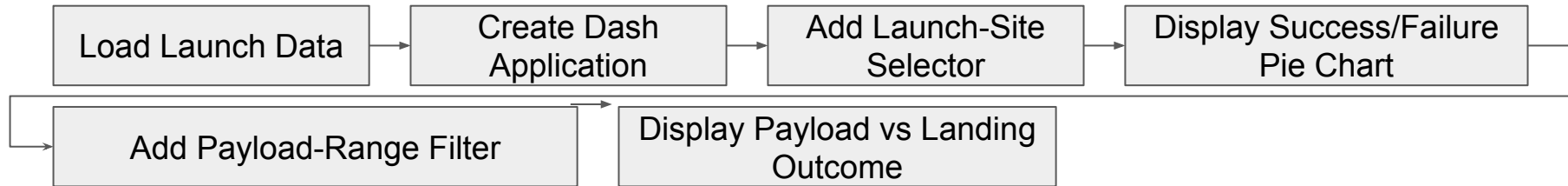
- Distance measured between launch site and coastline
- Calculated distance: 0.97 km

Interactive Visual Analytics: Plotly Dash - Methodology

Objective

Develop an interactive dashboard to explore Falcon 9 launch outcomes and payload relationships

Process



Dashboard Components

- Launch-site dropdown - All sites and individual launch sites
- Success-rate pie chart - View landing outcomes
- Payload-mass range slider - Filter launches by payload mass
- Payload mass vs. landing outcome scatter plot - Compare payload mass with landing class

Interactive Visual Analytics: Plotly Dash - Dashboard

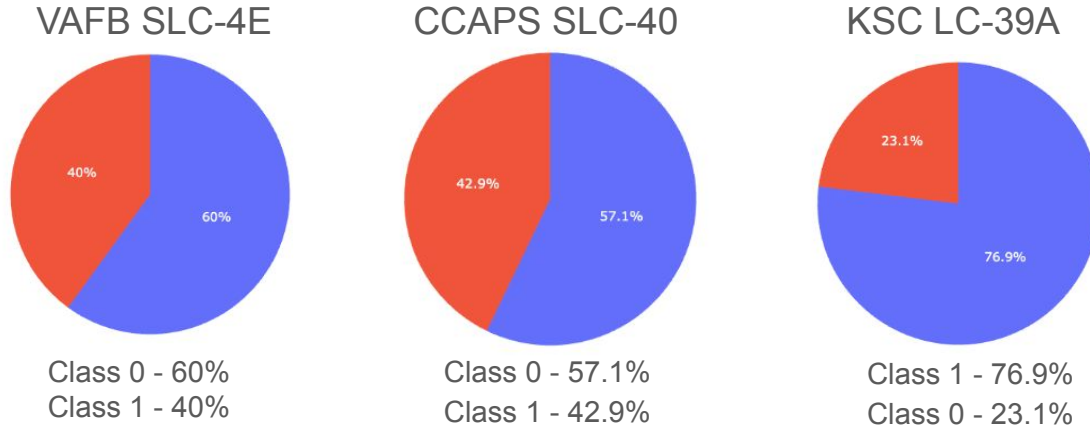
SpaceX Launch Records Dashboard



Dashboard Interface

- Launch-site selection allows results to be viewed across all sites or individual sites
- Payload range filtering allows the displayed launches to be restricted by payload mass
- Interactive charts update based on the selected filters

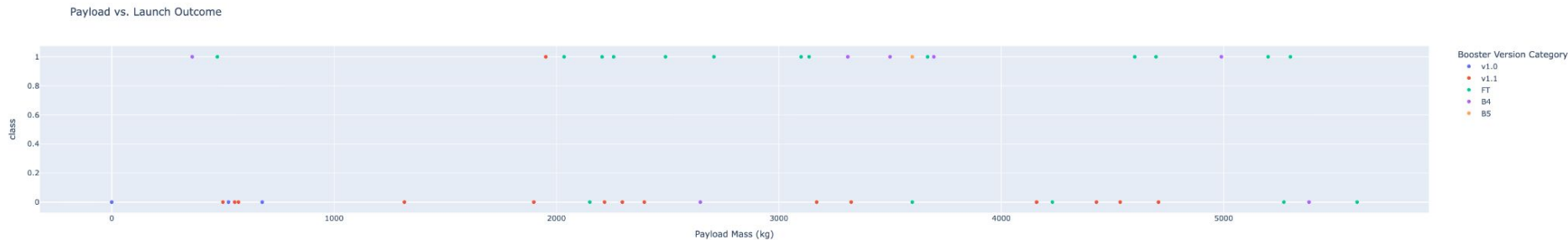
Plotly Dash Results: Launch-Site Outcomes



Launch-Site Outcome Comparison

- Landing outcomes differ across the selected launch sites
- KSC LC-39A has the largest Class 1 proportion among these views
- VAFB SLC-4E has the largest Class 0 proportion among these views

Plotly Dash Results: Payload vs. Landing Outcome



Payload and Landing Outcome

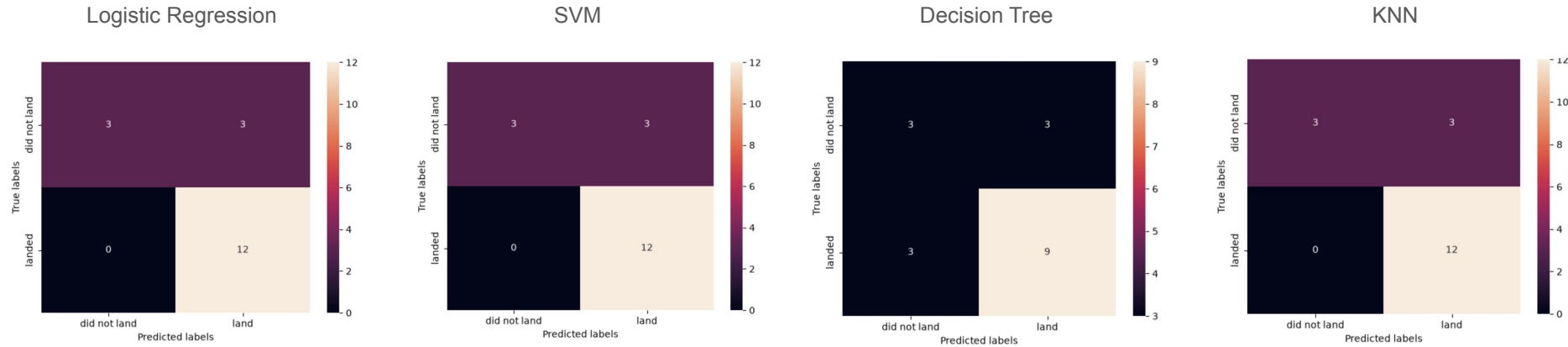
- Both landing classes are represented across the selected 0–6,000 kg payload range
- Successful and unsuccessful outcomes overlap across the payload range, with no clear separation by payload mass alone
- Multiple booster-version categories are represented among both landing outcomes

Predictive Analysis: Confusion Matrix Results

Objective

Examine correct and incorrect classifications produced by each machine-learning model

Confusion Matrix Results



Interpretation

The confusion matrices show the distribution of correct and incorrect classifications, providing more detail than overall accuracy alone

Predictive Analysis: Model Evaluation

Objective

Predict Falcon 9 first-stage landing success using historical launch characteristics.

Method



Models Evaluated

Logistic Regression <u>83.33%</u>	SVM <u>83.33%</u>	Decision Tree <u>66.67%</u>	KNN <u>83.33%</u>
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Total set: 18 observations

Key Observation

Logistic Regression, SVM, and KNN achieved 83.33% test accuracy, while Decision Tree achieved 66.67%.

Conclusion: Key Findings

Key Findings

- The project analyzed 90 Falcon 9 launches and their first-stage landing outcomes.
- Landing outcomes varied across launch sites, payload masses, orbit types, and time periods.
- Folium provided a geographic view of launch sites and landing outcomes, while Plotly Dash enabled interactive exploration of launch-site and payload relationships.
- Logistic Regression, SVM, and KNN achieved 83.33% test accuracy, while Decision Tree achieved 66.67%.

Overall Conclusion

Historical launch characteristics contain patterns associated with Falcon 9 first-stage landing outcomes, and machine-learning models can be used to classify landing success from these characteristics

Limitations

- Analysis was based on the available historical launch dataset.
- Model evaluation used a relatively small test set.
- Observed relationships do not establish causation.

Project Repository

<https://github.com/nittin-shankar/ibm-data-science-capstone>

Contains:

- Data collection notebooks
- Web scraping
- Data wrangling
- EDA and SQL analysis
- Folium visualization
- Plotly Dash application
- Predictive analysis